University of Pisa

Computer Engineering

Resource allocation in satellite communications

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Contents

Chapter 1

Introduction

The project consists of analyze the performance of a communication system with N terminals that send packets to a Ground Station (GS) via satellite, which acts as a relay node, following the specifications provided. Each terminal generates packets of size S bytes, which is a uniform distributed IID random variable between 4 and 100, every T seconds, where T is an exponentially distributed IID random variable with mean value of 2×10^{-3} s. Every 80×10^{-3} s a transmission frame, during which each active terminal (i.e., the ones that have at least one packet to send) generates a bearer index B between 2, 4, 8 and 16, and transmits to the satellite (otherwise it will transmit -1). The satellite forwards everything to the GS, which uses a priority queue to collect all the values received from the terminals (discarding the -1 ones) and serves them according to the highest bearer index first (if two or more packets have the same bearer index, they are served in FIFO order). The GS sends a grant to each terminal that can transmit and, after receiving the grant, the terminal sends up to M bytes, where M is based on B as follows:

$$M = 100 \times K^{\log_2(B)-1} \tag{1.1}$$

Chapter 2

Workflow

2.1 Define the objectives

We had to evaluate the performance of the overall system in terms of throughput for various values of N and K, with B uniformly distributed, as the specifications provided. We decided to analyze another metric, very important in communication systems: the average queue length at the terminals. To collect this data, we added an *Oracle* module to which each terminal sends the total bytes sent and the size of its queue length after every time frame.

Chapter 3

Results

Chapter 4

Conclusion